# Radio-frequency system and hardware modeling

## Performance Analysis of a realistic Cloud Radio Access Network (C-RAN)

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# 1 System Simulation

# Power Amplifier Characterization

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| Term |
| --- |
| Definition |

# LNA Measurement

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| Term |
| --- |
| Definition |

# 2 Hardware Experiments

# Power Amplifier Measurements

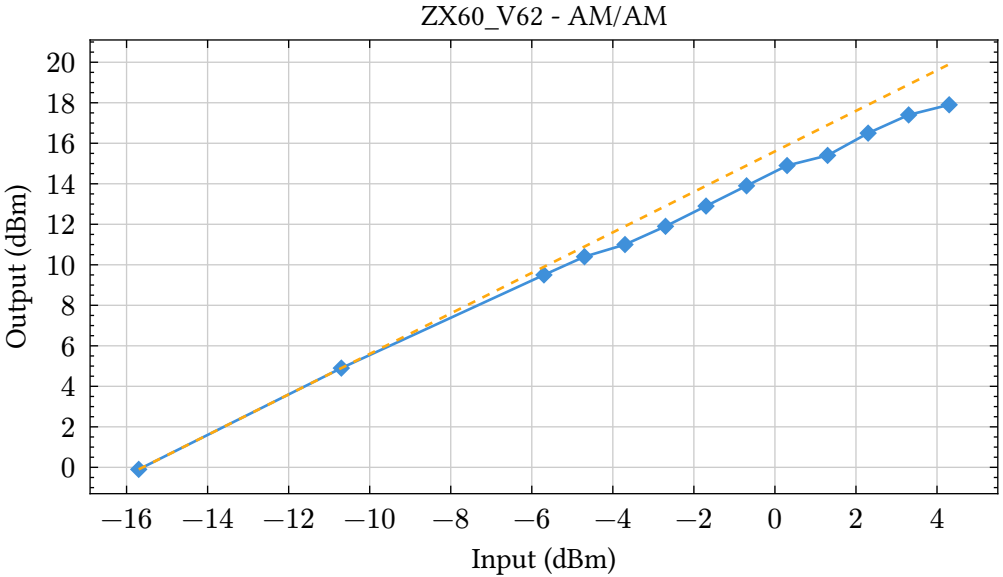



Figure 1: Curve of the measured AM/AM for the ZX60_V62 PA.

# Power Amplifier Measurements (ii)

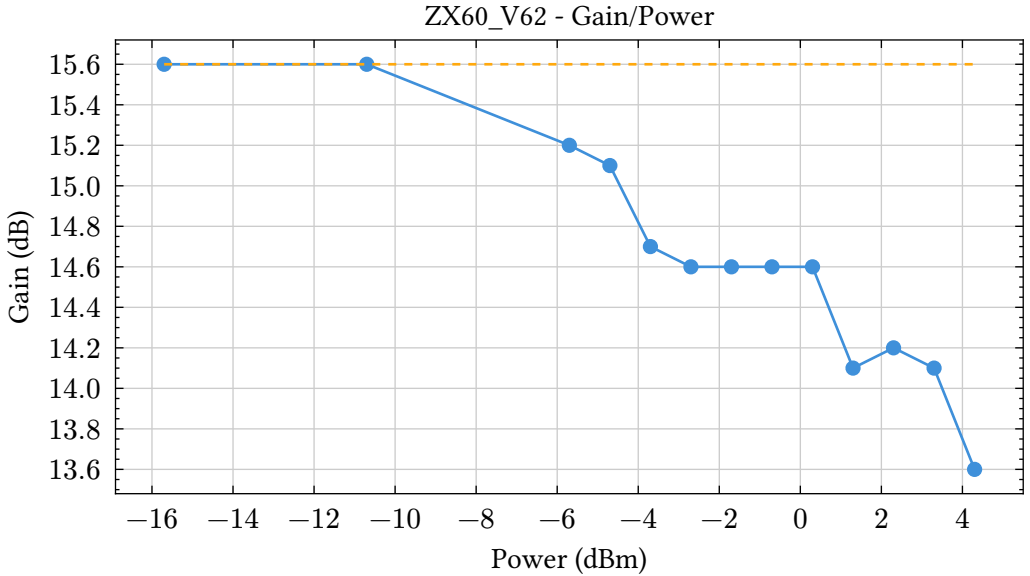



Figure 2: Curve of the measured Gain/Power for the ZX60_V62 PA.

# Power Amplifier Measurements (iii)

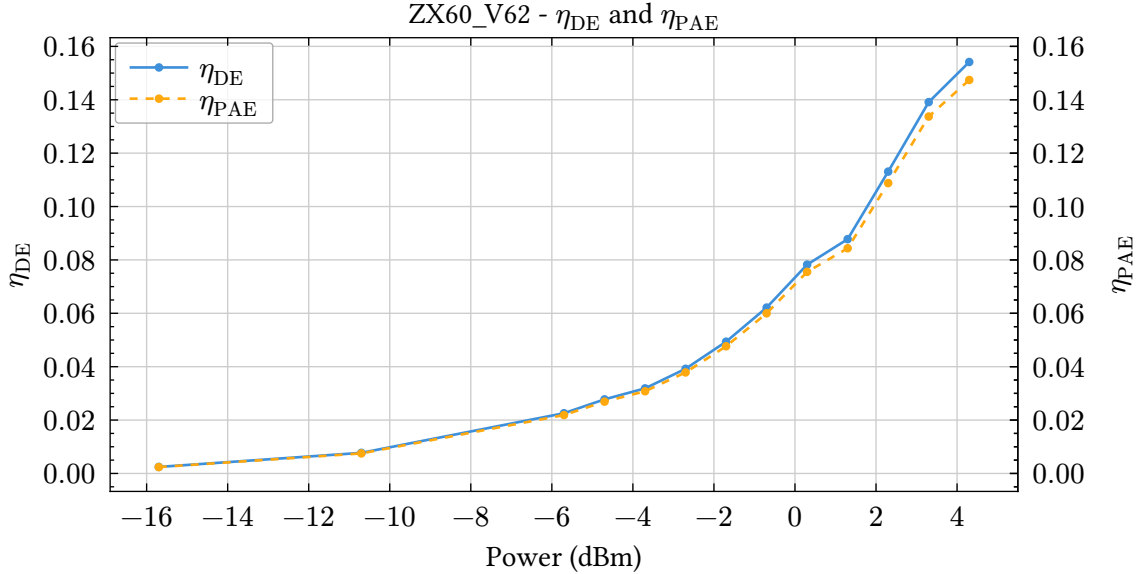



Figure 3: Curve of the measured $\eta_{DE}$ and $\eta_{PAE}$ for the ZX60_V62 PA.

# Power Amplifier Measurements (iv)

| Performances | Value | Unit |
|---|---|---|
| Gain | 15.6 | dB |
| $P_{1\text{dB}}$ | 12 | dBm |
| $V$ | 5 | V |
| $I$ | 89 | mA |
| $P$ | 0.445 | W |

Table 1: Measured

| Performances | Value | Unit |
|---|---|---|
| Gain | 15.4 | dB |
| $P_{1\text{dB}}$ | 19 | dBm |
| $V$ | 5 | V |
| $I$ | 82 | mA |
| $P$ | 0.41 | W |

Table 2: Datasheet

# Pluto Self-Communication

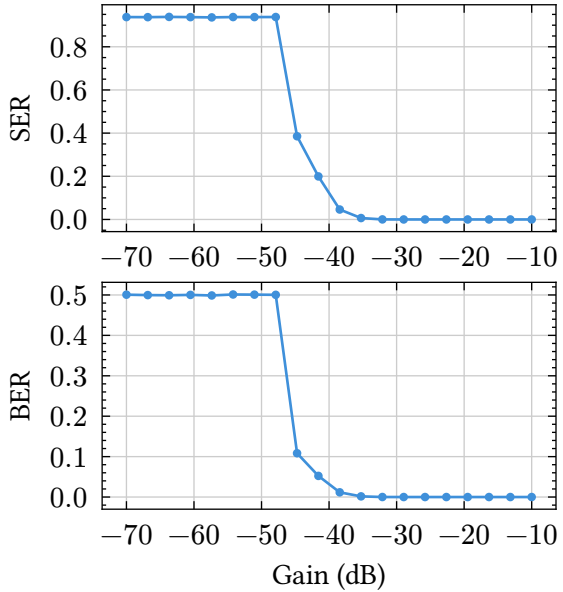



Figure 4: Full Scale at $-10$ dBFS (and $G_{\mathrm{RX}} = -10$ dB).

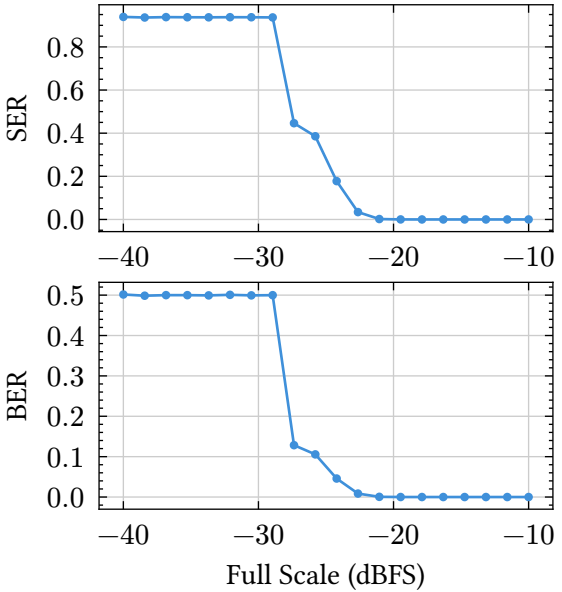



Figure 5: Fixed Gain at $-10$ dB (and $G_{\mathrm{RX}} = -10$ dB).

# Pluto Communication Demos

- Wired Single Pluto Communication Demo
- Wireless Pluto to Pluto Communication

# TX Pre Processing

1. Preamble addition
   - Barker Sequence
     - Sequence with high autocorrelation
       - Makes it easy to find from random symbols in RX
2. RRC Filter
   - Upsampling of the signal
   - Ensures the symbols don't "mix" with each other (reduce ISI)

# RX Post Processing

1. Coarse Frequency Compensation
   - Operates over the whole symbol stream
   - Helps with differences in the LO frequencies
   - Tries to make the symbols fit into the expected constellation type (QAM)
   - Internal PLL with PID
   - Does **not** decimate the signal
2. RRC Filter
   - Operates over the whole symbol stream after it is compensated
   - Helps to shape the symbols back to what they are supposed to be
   - Might have bad output from unaligned symbols
     - Loop over the different possible delays to ensure we use the best one
   - Decimates the signal
3. Carrier Synchronization (Fine Frequency Compensation)
   - Operates over the RRC decimated signal
   - Helps to compensate what the coarse compensation might have missed
   - Tries to make the symbols fit into the expected constellation type (QAM)

# RX Post Processing (ii)

- Internal PLL with PID
- Does **not** decimate the signal

4. Symbol Synchronization
   - Operates over the fine compensated symbols
   - Attempts even further to shape the symbols into the constellation
   - **Decimates** the signal
5. Preamble Detection
   - Helps us find the start of the message
   - Helps us to align the message in the time domain
   - The phase offset in the preamble tells us by how much the constellation is “rotated”